

Self-contained referee package – Reading-H vacuum selection over the

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This is a self-contained referee document for one result: within the certified operating region, the Reading-H vacuum selection of TECT prefers the isotropic Gaussian-Hartree layer G_* over every admissible competitor (lattice and non-lattice), given the A1 microscopic kernel convention. It collects every definition, lemma, and numerical check needed to verify or attack the result without consulting other files.

Claimed (tier T7-SCOPE, $\mathcal{C}_{\text{full}}$, comfortable off-diagonal):

$$F[Q] - F[G_*] > 0 \quad \forall Q \in \mathcal{C}_{\text{full}} \setminus \{G_*\}, \quad (I, \mu^2) \in \mathcal{R}, \quad \text{given A1.}$$

Not claimed: no statement is made for intensities outside $\mathcal{R}$, for the A1 convention being forced rather than chosen, or for a fully unconditional / all-competitor T7.

2. The object

TECT's vacuum-selection functional, restricted to the diagonal Gaussian-Hartree layer, assigns to a momentum-space condensate pattern Q (a finite point set on the Brazovskii shell $|k| = q_0$, with amplitudes) a free energy $F[Q]$. The reference G_* is the rotation-invariant (isotropic) dressing on the shell. The A1 kernel convention fixes the microscopic theory: $r_{\text{braz}} = K(q_0) = \mu^2$, with the Lagrangian constants $u = -0.86$, $v = 3.24$, $q_0 = 0.6802$, the gap point $r_R = r_R(\mu^2)$, and M_R the dressed shell mass.

Admissible class $\mathcal{C}_{\text{full}}$. A competitor Q is admissible iff it is (i) real antipodal, $Q = -Q$; (ii) packing-bounded, $|Q| \leq n_{\text{pack}} = 16/\theta_{\text{min}}^2$; (iii) coherence-resolved, any two points geodesically separated by $\geq \theta_{\text{min}}$, where $\theta_{\text{min}} = \sqrt{\hat{r}}/(2q_0^2\sqrt{c})$ is the coherent angular resolution ($\hat{r} = r_R + 2\lambda I$, $\lambda = 3u + 30vM_R$, c the gradient stiffness). Crystallographic (lattice) shells and arbitrary non-lattice configurations are both admitted. At the operating anchor $(I, \mu^2) = (2 \times 10^{-3}, 0.005)$, $\theta_{\text{min}} = 0.627$ and $n_{\text{pack}} = 40.68$.

Operating region $\mathcal{R}$ (Sec. 7): $\mu^2 \in (0, \mu_{\text{max}}^2)$, $I \in (0, I_c^{\text{sel}}(\mu^2))$.

3. The decomposition

Write $F[Q] - F[G_*]$ as the second variation at G_* plus higher cumulants, split into three sectors: (D) the diagonal (isotropy) Hessian; (O) the off-diagonal (condensate / additive) Hessian; (S) the third-cumulant selection floor (SC-SCOPE). The result follows from positivity of each sector over $\mathcal{C}_{\text{full}}$.

4. (D) Diagonal isotropy is competitor-agnostic

The second variation of the diagonal functional F_0 at G_* is

$$\delta^2 F_0 = \frac{1}{2} \int G_*(q)^{-2} (\delta G)^2 d^3 q + \Phi''_{\text{diag}} (\delta M_{\text{tot}})^2, \quad \Phi''_{\text{diag}} = \frac{3}{2} u_{\text{eff}},$$

with $u_{\text{eff}} = u + 10vM_R > 0$ and $M_{\text{tot}} = \int G$. Expanding an arbitrary perturbation $\delta G(q) = \sum_{lm} a_{lm}(|q|)Y_{lm}(\hat{q})$ and using $\int Y_{lm} d\Omega = 0$ for $l \geq 1$, the rotation-invariant interaction $\Phi_{\text{diag}}(M_{\text{tot}})$ curves only the $l = 0$ sector, so the Hessian is block-diagonal in angular momentum:

$$l = 0 : \frac{1}{2}G_*^{-2} + \frac{3}{2}u_{\text{eff}} > 0; \quad l \geq 1 : \frac{1}{2}G_*^{-2} \geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0,$$

the $l \geq 1$ infimum attained analytically at the gap shell $|q| = q_0$. The decomposition uses no lattice / additive structure: a non-lattice competitor is merely another direction δG , seen by the same positive block-diagonal Hessian. Hence the diagonal minimum is strict for every competitor.

5. (O) Off-diagonal: the elementary closure

The off-diagonal stability ratio is, with the additive-energy floor $K_{\text{floor}}(Q) = \sum_{t \neq 0} r(t)^2 / N^2 - 1$ ($r(t) = \#\{(a, b) \in Q^2 : a + b = t\}$, $N = |Q|$),

$$R_{\text{lead}}(Q) = \text{const} (1 + K_{\text{floor}}(Q)) I, \quad \text{const} = \frac{9}{4}u_{\text{eff}}^2 B_{\text{max}} \frac{2}{r_R} = 23.2,$$

N -independent, and the off-diagonal Hessian is positive iff $R_{\text{lead}} < 1$, i.e. $K_{\text{floor}} < K_* = 1/(\text{const } I) - 1 \simeq 20.5$. Two elementary lemmas bound K_{floor} .

Lemma 1 (additive-energy floor $\leq$ occupancy). $K_{\text{floor}}(Q) \leq T'(Q)$, where $T'(Q) = \max_{t \neq 0} \#(Q \cap C_t)$ is the maximal occupancy of a proper sum-circle $C_t = \{x : x \cdot t = |t|^2/2\} \cap S^2$. (This is the unconditional sum-circle bound $E_+(Q) \leq (1 + T')N^2$; both summands of $a + b = t$ lie on C_t .)

Lemma 2 (coherence circle-packing). For an admissible (coherence-resolved) Q and any $t \neq 0$, $\#(Q \cap C_t) \leq \lfloor 2\pi/\theta_{\min} \rfloor$. *Proof.* C_t is a Euclidean circle of radius $\rho = \sqrt{q_0^2 - |t|^2/4} \leq q_0$. Its Q -points are pairwise geodesically $\geq \theta_{\min}$, hence pairwise Euclidean $\geq d := 2q_0 \sin(\theta_{\min}/2)$. If k points lie on a circle of radius ρ with pairwise distance $\geq d$ then $2\rho \sin(\pi/k) \geq d$, i.e. $k \leq \pi / \arcsin(d/2\rho)$; since $\rho \leq q_0$, $d/2\rho \geq \sin(\theta_{\min}/2)$, so $\arcsin(d/2\rho) \geq \theta_{\min}/2$ and $k \leq 2\pi/\theta_{\min}$. ■

Across $\mathcal{R}$, $\theta_{\min} \geq 0.596$ so $2\pi/\theta_{\min} \leq 10.54$ and

$$K_{\text{floor}} \leq T' \leq 10 \Rightarrow R_{\text{lead}} \leq 23.2 \cdot 11 \cdot I \leq 0.615 < 1 \quad (\text{at } I = 2 \times 10^{-3} : R_{\text{lead}} \leq 0.510, \text{ margin } \times 1.96).$$

The off-diagonal cap $I_{\text{off}}^{\text{coh}} = 1/(23.2 \cdot 11) = 3.92 \times 10^{-3}$ exceeds the selection cap $I_c^{\text{sel}} = 2.41 \times 10^{-3}$, so the selection boundary binds and the operating region is the full Step-1 region (Sec. 7). The bound is uniform for lattice and non-lattice; a direct adversarial search finds the maximal admissible occupancy is $8 < 10$.

6. (S) Selection floor is competitor-agnostic

The SC-SCOPE third-cumulant survival is governed by the joint ratio $\text{joint} = \text{MARGIN}/(C_2 + \text{composed}/S_{\text{un}} + R_{\text{max}}C_2)$, built from layer quantities (the Prop-A margin MARGIN, the minimum-transfer envelope J_{eff} , the sunset cap $S_{\text{un}} = 1.13$, the realized quartic $R_{\text{max}} = 0.385$) and the coherence packing n_{pack} – the same for every admissible competitor, with no dependence on the competitor's additive structure. Numerically $\text{joint} = 1.04 > 1$ at the operating point, with the structural cap $\text{joint} \rightarrow S_{\text{un}} = 1.13$. Hence the selection floor is positive for non-lattice competitors identically.

7. Window certification

The Prop-A margin $\text{MARGIN}(\mu^2) = \text{PB}(M_+) - \text{DIP}$ requires the upper PD branch $M_+ = (-3u + \sqrt{9u^2 - 60v\mu^2})/(30v)$, i.e. $\mu^2 < \mu_{\text{max}}^2 = 3u^2/(20v) = 0.0342$. The selection critical intensity $I_c^{\text{sel}}(\mu^2)$ solves $\text{joint} = 1$; it is monotone increasing in μ^2 (implicit function theorem, with joint decreasing in I –

analytic, the denominator is linear-increasing in $C_2 \sim I^2$ with coefficient $1 + R_{\max} - 1/S_{\text{un}} = 0.500 > 0$ – and increasing in μ^2), with minimum $I_c^{\text{sel}} = 2.41 \times 10^{-3}$ on $[\times 0.5, \times 2]$. The certified region is

$$\mathcal{R} = \{(I, \mu^2) : 0 < \mu^2 < \mu_{\max}^2 = 0.0342, 0 < I < I_c^{\text{sel}}(\mu^2)\},$$

with the operating point $(2 \times 10^{-3}, 0.005)$ interior (20% intensity headroom).

8. The assembled result

(D) strict diagonal minimum (Sec. 4) + (O) $R_{\text{lead}} \leq 0.510 < 1$ comfortable (Sec. 5) + (S) joint > 1 (Sec. 6), all competitor-agnostic or coherence-bounded over $\mathcal{C}_{\text{full}}$, within the certified $\mathcal{R}$ (Sec. 7), give $F[Q] - F[G_*] > 0$ for every admissible competitor. The sole standing input is the A1 kernel convention (a definition of the microscopic theory, not a discharged-able hypothesis). Tier: T7-SCOPE on $\mathcal{C}_{\text{full}}$, comfortable off-diagonal margin.

9. Numerical reproduction

Each numerical claim is reproduced by a standalone script (sandbox, exit 0 iff all self-test asserts pass):

script	checks (expected: all PASS)
res5_032_window_certification.py	$I_c = 2.41 \times 10^{-3}$, $\mu_{\max}^2 = 0.0342$, monotonicity
res5_033_ext_adversarial_map.py	antipodal $T' \leq N/2$; $K_{\text{floor}} \sim 1\text{--}3$
res5_034_DS_nonlattice_extension.py	(D)/(S) competitor-agnostic
res5_035_cfull_scope_enactment.py	region, caps, margins
res5_036_coherence_offdiag_bound.py	$T' \leq 10$, $R_{\text{lead}} \leq 0.510$, selection binds

Run `python codes/vacuum/res5_03N_*.py`; each prints its asserts and exits 0.

10. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the coherence bound.* Lemma 2 rests on $\rho \leq q_0$ and pairwise $\geq \theta_{\min}$; both are definitional. A near-great sum-circle that could hold > 10 points forces its antipodal mirror within $< \theta_{\min}$ near the equator, violating admissibility (verified: maximal admissible occupancy 8). To falsify, exhibit an admissible Q with a sum-circle of > 10 points.
- (β) *Attack the A1 input.* The result is relative to the A1 kernel convention; it does not claim A1 is forced. A referee who rejects A1 rejects the theory’s definition, not this proof.
- (γ) *Attack the scope.* The claim is confined to $\mathcal{R}$ and to $\mathcal{C}_{\text{full}}$ (real antipodal, packing- and coherence-bounded). It is not a global or all-intensity statement; SC-SCOPE has a structural near-critical cap $S_{\text{un}} = 1.13$.

Quantitative sanity check (mandatory): $K_* = 20.55$; coherence bound $T' \leq 10$ (max admissible 8); $R_{\text{lead}} \leq 0.510$ ($\times 1.96$); $I_c^{\text{sel}} = 2.41 \times 10^{-3}$; $\mu_{\max}^2 = 0.0342$; entropy floor $4.64 \times 10^{-2} > 0$; joint $1.04 > 1$. All from res5_032–036.

11. Result footer

Result ID: Reading-H C-full referee package (self-contained)
Precise statement: Within $\mathcal{R} = \{\mu^2 \text{ in } (0, 0.0342), I \text{ in } (0, I_c^{\text{sel}}(\mu^2))\}$, given the A1 kernel convention, $F[Q] - F[G_*] > 0$ for every admissible competitor Q (real antipodal, $|Q| \leq n_{\text{pack}} = 16/\theta_{\min}^2$, pairwise geodesic $\geq \theta_{\min}$), lattice and non-lattice. Proof: (D) block-diagonal

isotropy Hessian (entropy floor $(1/2)r_R^2 > 0$, breathing $(3/2)u_{\text{eff}} > 0$), competitor-agnostic; (0) Lemma 1 ($K_{\text{floor}} \leq T'$) + Lemma 2 (coherence circle-packing $T' \leq \text{floor}(2\pi/\theta_{\text{min}}) = 10 \Rightarrow R_{\text{lead}} \leq 23.2 * 11 * I \leq 0.510 < 1$ (x1.96), selection binds; (S) joint=1.04>1, layer/packing, competitor-agnostic. Tier T7-SCOPE on C-full, comfortable off-diagonal. C-full competitor class, region R, given A1. NOT global/all-intensity/unconditional. Off-diagonal comfortable (x1.96); EXT (tight $K_{\text{floor}} - 3$) is an optional T2 further tightening, not used here.

Scope: A1 kernel convention (microscopic-theory definition); Lemma 1 (sum-circle bound); Lemma 2 (coherence circle-packing, elementary); res5_032/033/034/035/036.

Dependencies: REFERENCE PACKAGE (self-contained) + EXECUTED (res5_032--036 all PASS).

Evidence grade: python codes/vacuum/res5_032_window_certification.py ; res5_033_* ; res5_034_* ; res5_035_* ; res5_036_coherence_offdiag_bound.py

Reproduction command: each script prints its asserts and exits 0.

Expected output: an admissible Q with a sum-circle occupancy > 10 (breaks Lemma 2); or $F[Q] \leq F[G^*]$ for some admissible Q in R; or $\theta_{\text{min}} < 2\pi/11 = 0.571$ in R.

Falsifier: standalone assembly of the enacted T7-SCOPE_{C-full} (B1/B2); no tier change. External-referee package (validation/publication gate G2).

Tier before / after: Scope-qualified T7 of the Reading-H comparison over C-full, given A1, within R, comfortable margin. NOT a global / all-competitor / all-intensity / A1-free claim.

No-overclaim statement: external / second-author reproduction (run res5_032--036) and review of Lemmas 1--2 and the (D)/(S) competitor-agnosticism arguments.

Next required action: